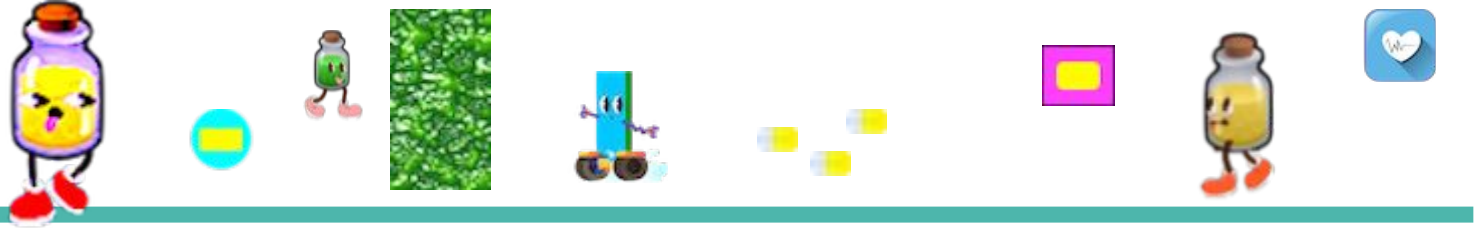

COMP 486 A3: Sudsy's Cleaning Survival

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Presentation Outline

PART 1: Game Idea and Design

PART 2: Improvements since Assignment 2 (A2)

- A. A2 Recap
- B. New Maps & Menus
- C. New HUD & Feedback
- D. New Weapons & Sounds
- E. New Enemies & Game Play Modifications
- F. New Game Cost Logic

PART 3: Future Improvements

PART 1: Game Idea and Design

Game Idea

- Sudsy Spongerton (a sponge) needs to protect their space from the chemicals trying to get them.
  - Fun play on a survival game.

- Motivation:
  - My goal to make the game fun, approachable, and play on different things a sponge runs into when cleaning around a home.

Game Design - Survival Game

- Built using Phaser and Tiled.
- Top-down view 2D.
  - Control player with arrow keys.
  - Space bar to shoot
- Choose the map you want to use.
  - The map will have areas blocked off by 'messes'.
- Enemies come in waves.
  - Destroy all enemies in the wave to reach the next level.
  - Waves progressively get larger.
- Get points for destroying enemies.
  - Points are used for:
    - Unlocking gates (i.e. 'cleaning messes').
      - Allow user to explore a larger area.
    - Purchasing weapons.
      - All weapons have ammo limits.

PART 2: Improvements Since A2

A. A2 Recap

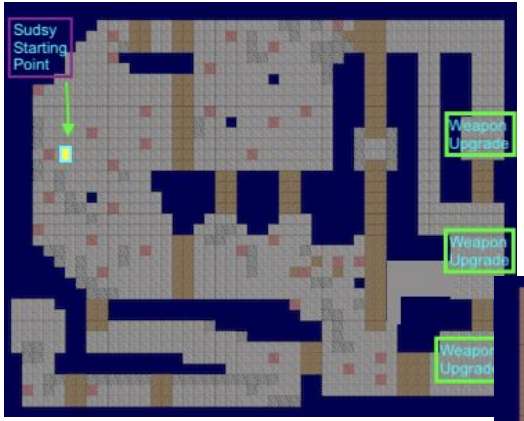
Assignment 2 Recap

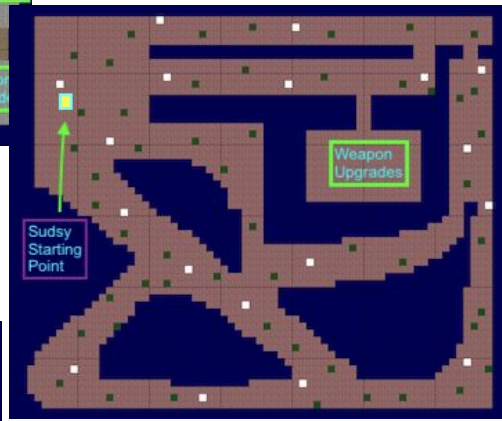
- One level
- Pickups: Health and Ammo
- Gates were the only purchasable item.
- Only one weapon.
- 2 types of enemies.
- Simple HUD
- Pause menu
- Simple Game Over Screen.
- No sound
- No restart - browser refresh only.
- Player could be pushed through the wall and gates.

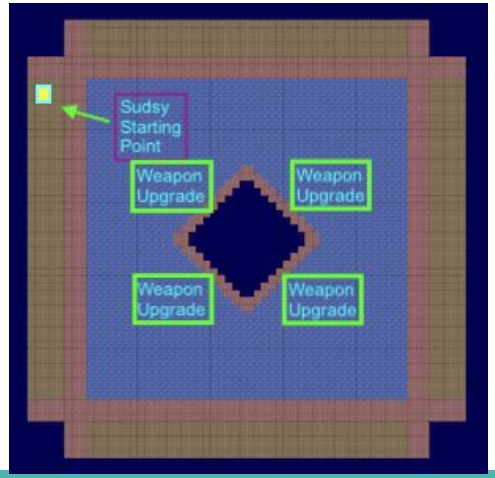
B. New Maps & Menus

A3 Additions - Maps

- 3 different maps (sponge themed)
- Built with different user goals in mind.
  - Kitchen Map
    - Built from the map of assignment 2.
    - Smaller, close quarters.
  - Drain Pipes Map
    - Medium in size.
    - Lots of exploration options.
  - Bathroom Map
    - Larger in size.
    - Immediate access to weapons (other levels required unlocking).
    - Long term survival.

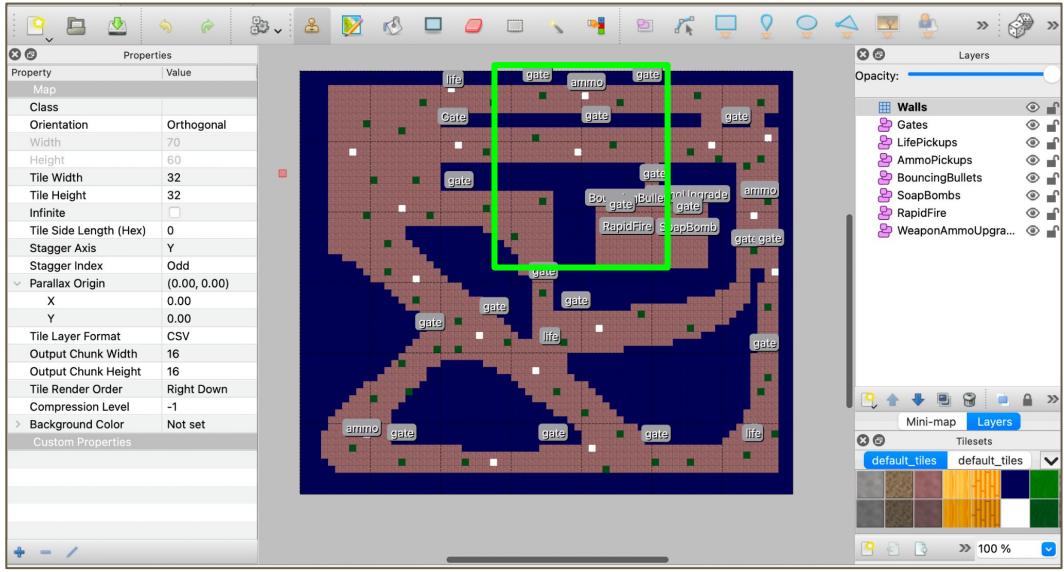






A3 Additions - Maps

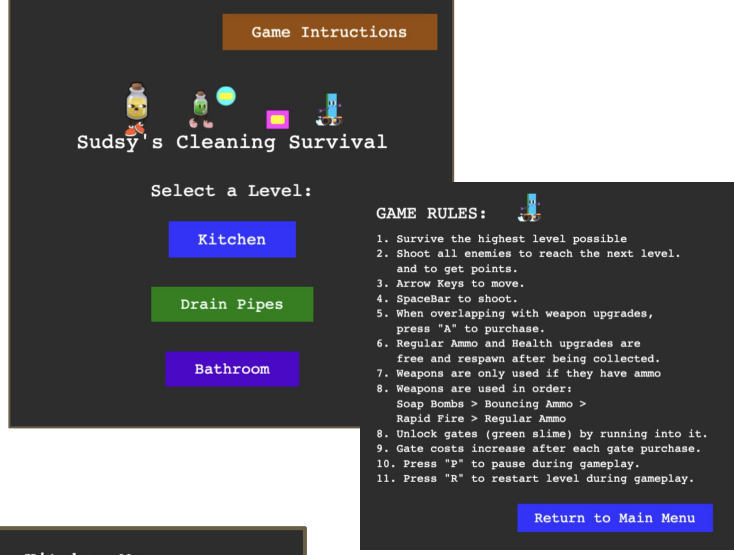
- Using Tiled:
  - Tiled = an app to create 2D maps.
    - Download as Json file and then loaded into the game world with Phaser.
  - For A2:
    - all upgrades and gates were placed by hand using x,y coordinates.
  - For A3:
    - Learned how to create different layers in Tiled app.
    - Then items were automatically drawn and click and drag to place them.

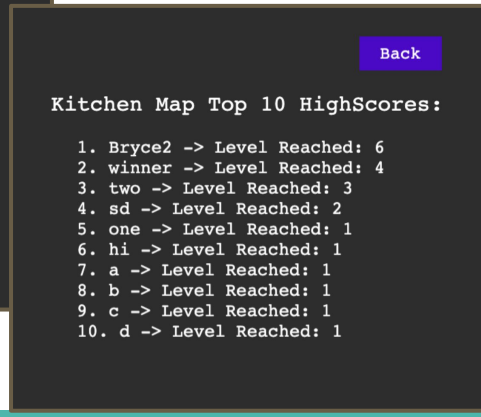


- Each map has a slightly different camera zoom.
  - Minimal, but I found the larger map, it felt better to be zoomed out a bit more.

A3 Additions - Menus + HighScores

- Main menu
  - All buttons in the game have a hover effect.
  - Required restructuring my code from A2 to have the main game scene logic separate from index.html.
- Game Instructions
  - An overview of all game instructions if needed.
- Map menus
  - Display map:
    - Name, description, image, and options.
  - High Scores Page:
    - Stored locally, but persistent (LocalStorage).
    - Keeps and displays only the top 10 scores.
    - Scores are ordered by the level reached.





A3 Additions - High Score Submission

- An input box added to the Game Over Screen.
- User can submit their score to be stored in the High Scores.
  - Enter name or it is saved as BLANK.
  - Name entered has a max limit of 10 characters.
- Only kept if Top 10.

1. Show in Game

C. New HUD & Feedback

A3 Additions - Start Screen + Pause Screen

- Start and pause screens are the same scene
  - Wording is appropriate for the instance
    - Start vs Resume
    - Map name vs Paused

- Game Instructions
  - The succinct game instructions are included as well.

A3 Additions - Pause and Restart

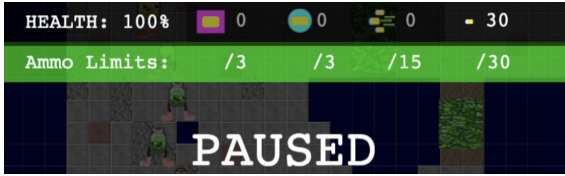
- Pause
  - Press 'P' during gameplay.
- Restart
  - Press 'R' during gameplay.
  - Press restart on Pause or GameOver Menu.

A3 Additions - Game HUD + User Feedback

- Ammo Limits Displayed in Pause Screen.
  - Update as ammo limits are increased.
- HUD includes health, all weapons, points, level reached.
- User Feedback:
  - When reach a new level, level text enlarges and blinks yellow.
  - Points blink red if cannot afford something.
  - Ammo blinks red if it is maxed out.
- HUD indicates which weapon is being used.
  - Weapons discussed on next slide.

D. New Weapons & Sounds

A3 Additions - Weapons/ upgrades



- Added purchasable Items (staying to theme of cleaning):
  - Soap Bombs
    - Place as a static object.
    - Destroys enemies when they run into it, up to 3 enemies destroyed.
  - Bouncing Bullets
    - Bounce off of the walls a total of 3 times.
    - Destroys all enemies it hits.
  - Rapid Fire Ammo
    - Increased firing rate of weapons.
  - Ammo Limit Increase
    - Increases the ammo limit of each item.
    - This limit is presented in the pause menu.
- Weapons are used in a priority order:
  - Bombs > Bouncing > Rapid > Regular
  - Chosen as bombs can then be strategically placed and the rest remain for beginning next level.
  - Limited ammo amounts.
- Press A to purchase.

A3 Additions - Sounds

- Rationale for sounds:
  - I started playing the game and decided where I felt I needed more input and where it was distracting.
- Added sounds for:
  - Shooting
  - Purchases
  - Collision with enemies
  - Gates collision and unlocking
  - Maxing out ammo
  - Empty ammo
- Removed sounds for:
  - Destroying enemies (it was too distracting)

2/3. Show in Game

E. New Enemies & Game Play Modifications

A3 Additions - Enemies + Enemy display

- 4 Enemies (I built all sprites):
  - All enemies can walk through green messes.
  - Small Enemy - Same as A2
    - Destroyed with 1 shot.
  - Fast Enemy
    - Moves faster.
    - 1 shot to destroy.
    - Small enemy tinted white.
  - Large Enemy
    - Destroyed with 2 shots.
    - Changes colour to show damage.
  - Largest "Boss" Enemy
    - Destroyed with 5 shots.
    - Changes colour to show damage.
    - Does not collide with walls or other enemies.
    - Walks a bit slower.

A3 Additions - Enemies + Enemy display

- Enemy waves
  - Number of waves matches the level
    - Level 1 = 1 wave
    - Level 2 = 2 waves
    - etc.
  - Odd Waves
    - 5 small enemies
    - 1 larger enemy
  - Even Waves
    - 1 Boss enemy
    - 4 small enemies
    - 1 fast enemy

A3 Additions - Gate Text

- Made text more legible (bold with stroke surrounding).

A3 Additions - Slightly longer life, Player collisions

- Health
  - Enemy collisions are slightly less damaging to allow for slightly longer life.
  - With the larger enemies, the player was dying very quickly.
- Player collisions
  - Previously the enemies could push player back.
    - Player could get trapped with large enemy.
    - Player could be pushed into the walls and through gates.
  - Player cannot be pushed anymore:
    - Does not get trapped.
    - Cannot drive through walls or gates.

4. Show in Game

F. New game cost logic

A3 Additions - Points + Cost Logic

<u>Points Received</u> (respective of number of shots needed to destroy them)

- Small Enemy - 10pts
- Fast Enemy - 10pts
- Large Enemy - 20pts
- Boss Enemy - 50pts

Enemy waves:

- Regular wave = 70 pts
- Every second wave = 100 pts

<u>Costs</u>

- Gates - cost starts at 100pts
  - Cost increases with each gate purchased. (100 + 50*(GatesOpened+1)).
  - Forces user to be strategic with gate opens.
  - Gates also become an object to avoid too.
- Soap Bomb - 40pts
  - Destroys 1-3 enemies (30-70pts depending on the enemies colliding).
- Bouncing Bullet - 80pts
  - Destroys unlimited number of enemies.
- Rapid Fire - 60pts
  - 15 Bullets (150 pts if perfectly shot)
- Ammo Limit Increase - 600pts
  - Needs to be more expensive or else makes game too easy.

Part 3: Future Improvements

Future Improvements

- Smart Enemies
  - Add in an enemy that will search level to find user.
- More levels and weapons
  - Add in teleportation, maybe 3D jumping effects, more weapon choices etc.
- Long term gameplay incentive
  - Having to navigate a maze to get to a goal.
  - As I tried to do, build maps that have different playing styles in mind for variability.
  - Add in optional mini games to break up game play.
- More purchasable items
  - Walls to build
  - Turrets to place
  - Companions to fight enemies
- Modifications and additions are endless.

End :)

Thank you for listening.

Any Questions?